



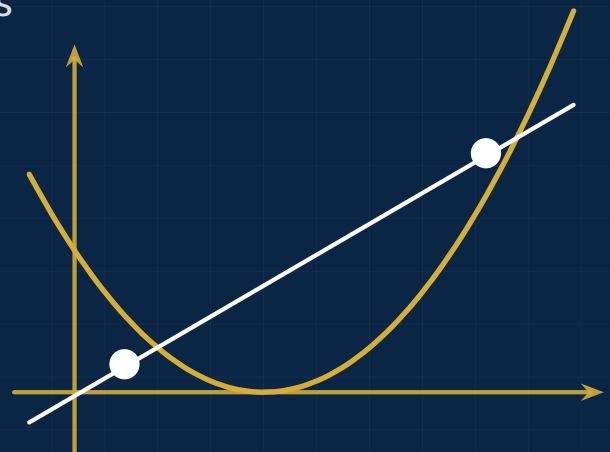
DR ESLAM AHMED  
eliteigcse.com

CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 4

# Linear Graphs

$$y = mx + c$$

$m$  is the steepness,  $c$  is where it crosses the  $y$ -axis. Read them, draw them, find them — and flip  $m$  for a perpendicular.



5

VARIANTS

7

PAST-PAPER PARTS

58

PRACTICE QUESTIONS

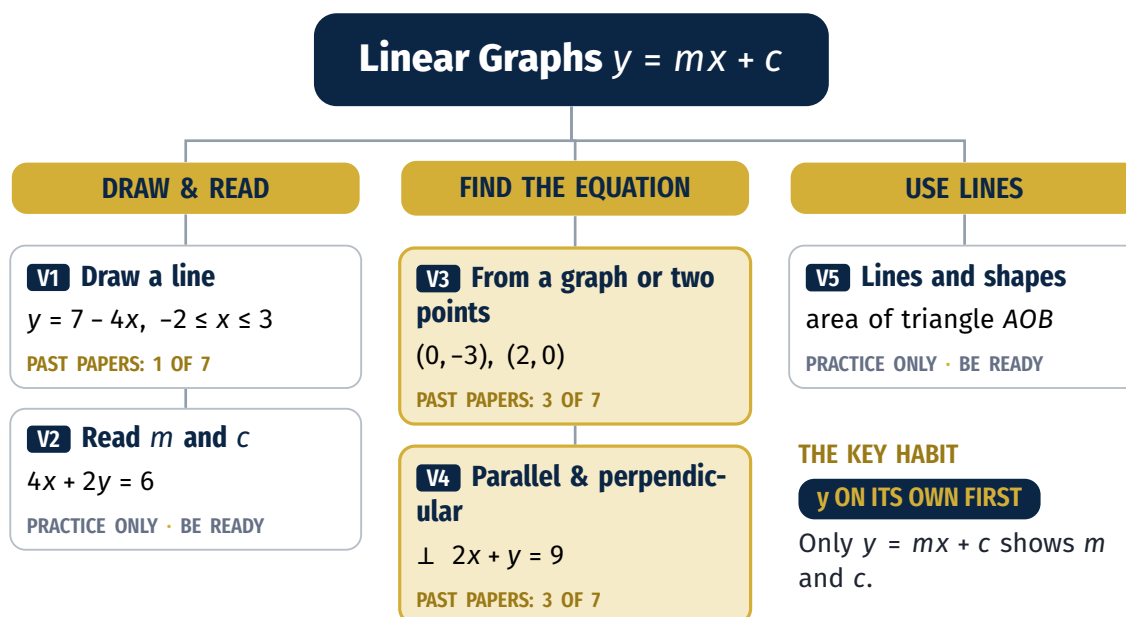
## YOU WILL MASTER

V1 Draw a line   V2 Read  $m$  and  $c$    V3 Equation from a graph

V4 Parallel & perpendicular   V5 Lines and shapes

## TOPIC MAP · EVERY VARIANT ON ONE PAGE

# Linear graphs at a glance


  
Your notes


## ★ How often does each variant come up?

7 past-paper questions (Oct 2020 → Nov 2024); 58 practice parts — grouped by type. Harder line work is in Topic 2 (Coordinate Geometry).

|                                    | PAST PAPERS (OUT OF 7) |   | PRACTICE PARTS |  |
|------------------------------------|------------------------|---|----------------|--|
| <b>V1</b> Draw a line              | ● ● ● ● ● ● ●          | 1 | 7              |  |
| <b>V2</b> Read $m$ and $c$         | ● ● ● ● ● ● ●          | 0 | 8              |  |
| <b>V3</b> From a graph, two points | ● ● ● ● ● ● ●          | 3 | 11             |  |
| <b>V4</b> Parallel, perpendicular  | ● ● ● ● ● ● ●          | 3 | 18             |  |
| <b>V5</b> Lines and shapes         | ● ● ● ● ● ● ●          | 0 | 14             |  |

**WHAT THIS TELLS YOU** Short, early questions (Q3–Q18, 2–4 marks): draw a line, read its equation off a grid, or a parallel/perpendicular line. Quick marks once  $y = mx + c$  is automatic.

## BEFORE YOU START

## The big idea

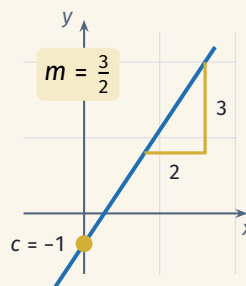


$y = mx + c$  tells you two things.

$m$  — the gradient: how far up for 1 across (negative: downhill).

$c$  — the  $y$ -intercept: where the line crosses the  $y$ -axis, at  $(0, c)$ .

Every point  $(x, y)$  on the line makes the equation true.



## Four facts

## Gradient

$$m = \frac{\text{rise}}{\text{run}}$$

Up is +, down is -.

 $y$ -intercept

$$(0, c)$$

Put  $x = 0$ .

 $x$ -intercept

$$y = 0$$

Solve for  $x$ .

## On the line?

substitute

Both sides equal:  
yes.

## Related lines

## Parallel

same  $m$

$$y = 2 - 5x \text{ and } y = 6 - 5x.$$

## Perpendicular

$$m \rightarrow -\frac{1}{m}$$

$$-2 \rightarrow \frac{1}{2}.$$

## Rearranged

$$5y + 2x = 7$$

$$y = -\frac{2}{5}x + \frac{7}{5} \text{ first.}$$



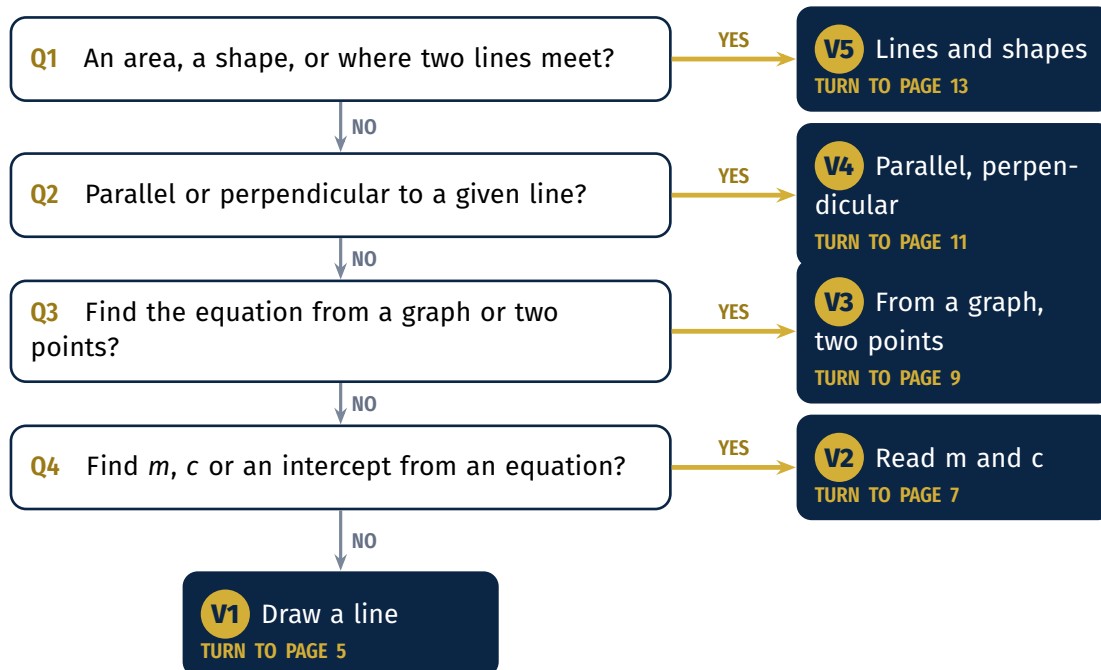
## CALCULATOR FIRST, THEN CHECK

Calculator table mode ( $f(x) = 7 - 4x$ , start  $-2$ , end  $3$ , step  $1$ ) gives the whole table in one go — then plot.

COUNT / RECOGNISE, THEN CHOOSE

# Which variant am I?

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



## COLOUR CODE IN THE WORKING

**Blue** = letter terms  $4x$ ,  $-3y$

**Orange** = plain numbers  $17$ ,  $-13$

**STEP 1** **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

## ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

Your notes

**V1**

VARIANT 1

# Draw a Line from its Equation

Table of values, plot every point, join with one ruled line.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● 1/7

PRACTICE QUESTIONS: 7



Your notes

Came up in: Oct/Nov 2020 P1H Q3

## ? What is it?

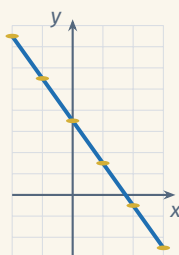
### In plain words

Make a **table**: every  $x$  in the range, work out  $y$ . Plot the points — they must lie in a **straight line**.

Rule **one line** through all of them, across the whole range asked.

### IT LOOKS LIKE

$$y = 7 - 4x \text{ for } -2 \leq x \leq 3$$



|     |    |    |   |   |    |    |
|-----|----|----|---|---|----|----|
| $x$ | -2 | -1 | 0 | 1 | 2  | 3  |
| $y$ | 15 | 11 | 7 | 3 | -1 | -5 |

## How do I do it?

### 1 Table

Every whole  $x$  in the range.

### 2 Work out $y$

Careful with negatives:  $7 - 4(-2) = 15$ .

### 3 Plot

Small crosses, exactly on the grid.

### 4 Join

One ruled straight line, end to end.

**TABLE, PLOT, RULE** A point off the line = a slip in the table.

**V1 Draw a Line from its Equation**

VARIANT 1 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"Draw the graph of  $y = 7 - 4x$  for values of  $x$  from  $-2$  to  $3$ " → **Table  $x = -2, \dots, 3$ ; plot; rule**

"Complete the table of values" → **Substitute each  $x$**

A point that is not in line → **Recheck that  $y$  value**

**WORKED EXAMPLE** **PAST PAPER**

Oct/Nov 2020 · Paper 1H · Q3 · 3 marks

On the grid, draw the graph of  $y = 7 - 4x$  for values of  $x$  from  $-2$  to  $3$ .

**STEP 1**  $x : -2, -1, 0, 1, 2, 3$  ◀ the range

**STEP 2**  $y : 15, 11, 7, 3, -1, -5$  ◀  $7 - 4x$  each time

**STEP 3**  $(-2, 15), (-1, 11), \dots, (3, -5)$  ◀ plot

**STEP 4** One ruled line from  $(-2, 15)$  to  $(3, -5)$  ◀ join

**ANSWER** straight line through  $(-2, 15)$  and  $(3, -5)$  ✓  $m = -4$ : down 4 for each 1 across ✓

**WATCH OUT**

$$\times 7 - 4(-2) = -1$$

$$\checkmark 7 - 4(-2) = 7 + 8 = 15$$

Minus times minus is plus.

**EXAM TIP**

Draw the line across the whole range — stopping short loses the mark.

**Check yourself**

calculator first · answers on page 15

**1.1**

Dr Eslam

Complete the table for  
 $y = 2x - 1$ ,  
 $x = -1, 0, 1, 2, 3$ .

Answer: .....

**1.2**

Practice · Easy

Complete the table for  
 $y = 2x + 5$ ,  
 $x = -2, -1, 0, 1, 2$ .

Answer: .....

**1.3**

Practice · Easy

$y = 8 - 2x$  for  $-1 \leq x \leq 4$ .  
Give the end points of  
the line.

Answer: .....



**V2** Read  $m$  and  $c$ 

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 **WHEN YOU SEE → YOU DO**

"Find the gradient of  $4x + 2y = 6$ " →  $y = -2x + 3$ ;  $m = -2$

"Where does the line cross the  $y$ -axis?" →  $(0, c)$

"Show that the lines are parallel" → **Rearrange both; same  $m$**

**WORKED EXAMPLE** DR ESLAMRead  $m$  and  $c$  · 2 marksFind the gradient and the  $y$ -intercept of the line  $4x + 2y = 6$ .**STEP 1**

$2y = -4x + 6$

◀  $x$  term across**STEP 2**

$y = -2x + 3$

◀ divide every term by 2

**STEP 3**

$m = -2, \quad c = 3$

◀ read

**ANSWER** gradient  $-2$ , crosses at  $(0, 3)$ ✓  $x = 0$ :  $2y = 6$ ,  $y = 3$  ✓**WATCH OUT**✗  $4x + 2y = 6$ :  $m = 4$ ✓  $m = -2$ 

$y$  is not on its own, so 4 is not the gradient.

**EXAM TIP**

"Parallel?" — rearrange both lines and write both gradients side by side.

**Check yourself**

calculator first · answers on page 15

**2.1**

Dr Eslam

Find the gradient and  $y$ -intercept of  $3x - 2y = 4$ .

Answer: .....

**2.2**

Dr Eslam

Where does  $y = 4x - 8$  cross the  $x$ -axis?

Answer: .....

**2.3**

Practice · Medium

Are  $3y = 9x - 5$  and  $y = 3x + 2$  parallel?

Answer: .....



**V3****VARIANT 3** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

●●●●●●● 3/7

PRACTICE QUESTIONS: 11

# Equation from a Graph or Two Points

Read  $c$  where it crosses the  $y$ -axis; rise over run for  $m$ ; write  $y = mx + c$ .

**Came up in:** Jun 2022 P1HR Q10 • Nov 2023 P2H Q9 • May 2024 P1HR Q9

 Your notes

## ? What is it?

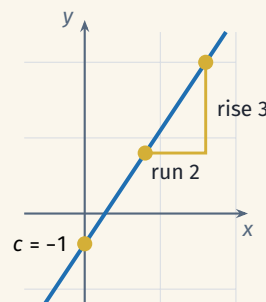
### In plain words

From a graph:  $c$  is where the line crosses the  $y$ -axis. Pick two points **exactly on grid corners**;  $m = \frac{\text{rise}}{\text{run}}$ .

From two points:  $m$  first, then  $c$  (or  $y - y_1 = m(x - x_1)$ ).

### IT LOOKS LIKE

line on a grid     $(0, -3)$  and  $(2, 0)$



## How do I do it?

**1**  $c$ Where it crosses the  $y$ -axis.**2** Two corners

Points exactly on the grid.

**3**  $m$ 

Rise over run; downhill is negative.

**4** Write $y = mx + c$ ; check a third point. **$c$  FIRST, THEN RISE OVER RUN**

Use the axis scales, not squares.

**V3 Equation from a Graph or Two Points**

VARIANT 3 · SEE IT DONE, THEN TRY

  
 Your notes
**WHEN YOU SEE → YOU DO**

"Line L is drawn on the grid. Find an equation for L"

→ **c from the axis; rise/run**

"passes through (0, -3) and (2, 0)"

→  $c = -3; m = \frac{3}{2}$

"in the form  $y = mx + c$ "

→ **Fractions allowed:**  $y = -\frac{1}{2}x + 1$

**WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 1HR · Q10 · 3 marks

Line L is drawn on the grid. It crosses the y-axis at (0, -1) and passes through (2, 2) and (4, 5). Find an equation for L in the form  $y = mx + c$ .

**STEP 1**  $c = -1$

◀ crosses the y-axis

**STEP 2** (2, 2) → (4, 5)

◀ two grid corners

**STEP 3**  $m = \frac{5-2}{4-2} = \frac{3}{2}$

◀ rise over run

**STEP 4**  $y = 1.5x - 1$

◀ write it

**ANSWER**  $y = 1.5x - 1$

✓ (4, 5):  $6 - 1 = 5$  ✓

**WATCH OUT**

× counting squares when the scale is 2

✓ use the axis numbers

The scale decides the rise and run, not the squares.

**EXAM TIP**

A line going down left to right must have a negative  $m$  — check the sign.

**Check yourself**

calculator first · answers on page 15

**3.1**

May 2024 P1HR

Line L crosses the y-axis at (0, 1) and the x-axis at (2, 0). Find  $y = mx + c$ .

Answer: .....

**3.2**

Nov 2023 P2H

A line passes through (0, -3) and (2, 0). Find an equation of the line.

Answer: .....

**3.3**

Dr Eslam

Find the equation of the line through (2, 5) and (6, 13).

Answer: .....

**V4****VARIANT 4** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

●●●●●●●● **3/7**

PRACTICE QUESTIONS: 18

# Parallel & Perpendicular Lines

Parallel: same  $m$ . Perpendicular:  $-\frac{1}{m}$ . Then use the point to find  $c$ .

**Came up in:** Jun 2024 P2H Q18 • Nov 2024 P2H Q18 • Jun 2025 P2HR Q10

 Your notes

## ? What is it?

### In plain words

Get the gradient of the given line ( **$y$  alone first**). Parallel: **same**.

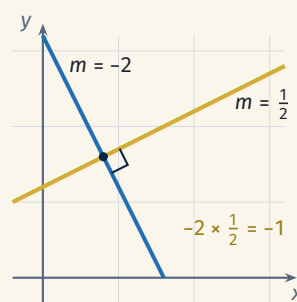
Perpendicular: **flip and change sign**.

Put the given point into  $y = mx + c$  to find  $c$ .

### IT LOOKS LIKE

$$\perp 2x + y = 9 \text{ through } (8, 11)$$

$$\parallel y = 2 - 5x$$



## How do I do it?

### 1 Given $m$

Rearrange the given line.

### 2 New $m$

Same, or  $-\frac{1}{m}$ .

### 3 Find $c$

Substitute the point.

### 4 Write

In the form asked.

**FLIP AND CHANGE SIGN**  $-2 \rightarrow \frac{1}{2}$ ,  $\frac{2}{5} \rightarrow -\frac{5}{2}$ .

## V4 Parallel & Perpendicular Lines

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

### WHEN YOU SEE → YOU DO

“parallel to  $y = 2 - 5x$  ... passes through  $(0, 6)$ ”

→  $m = -5, c = 6$

“perpendicular to  $2x + y = 9$ ”

→  $y = -2x + 9$ : **new**  $m = \frac{1}{2}$

“Find the gradient of a line perpendicular to P”

→ **Just**  $-\frac{1}{m}$  — **no equation needed**

### WORKED EXAMPLE PAST PAPER

November 2024 · Paper 2H · Q18 · 4 marks

The straight line **L** is perpendicular to  $2x + y = 9$  and passes through  $(8, 11)$ . Find an equation for **L** in the form  $y = mx + c$ .

**STEP 1**  $y = -2x + 9 \Rightarrow m = -2$

◀ given line,  $y$  alone

**STEP 2**  $m_L = \frac{1}{2}$

◀ flip and change sign

**STEP 3**  $11 = \frac{1}{2}(8) + c \Rightarrow c = 7$

◀ use the point

**STEP 4**  $y = \frac{1}{2}x + 7$

◀ write

**ANSWER**  $y = \frac{1}{2}x + 7$

✓  $-2 \times \frac{1}{2} = -1$  ✓

### WATCH OUT

×  $2x + y = 9$ :  $m = 2$

✓  $m = -2$

Rearrange first:  $y = -2x + 9$ .

### EXAM TIP

Check perpendicular gradients multiply to  $-1$ .

### Check yourself

calculator first · answers on page 15

4.1

Jun 2024 P2H

**P:**  $5y + 2x = 7$ . Find the gradient of a line perpendicular to **P**.

Answer: .....

4.2

Jun 2025 P2HR

**L** is parallel to  $y = 2 - 5x$  and passes through  $(0, 6)$ . Find **L**.

Answer: .....

4.3

Dr Eslam

Find the line perpendicular to  $y = 3x - 1$  through  $(6, 2)$ .

Answer: .....

**V5**

## VARIANT 5

## Lines and Shapes

Intercepts give triangle corners; two lines meet where their equations agree.

## SEEN IN PAST PAPERS

0/7

**NOT ASKED YET • BE READY**

## PRACTICE QUESTIONS: 14



Your notes

## ? What is it?

### In plain words

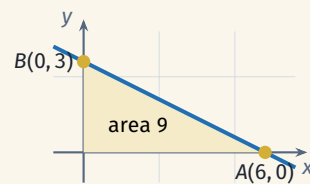
Where a line crosses the **axes**: put  $x = 0$ , then  $y = 0$ . With the origin these make a **right-angled triangle**:  
area =  $\frac{1}{2} \times \text{base} \times \text{height}$ .

Two lines **meet** where both equations are true: set them equal.

## IT LOOKS LIKE

area of  $AOB$

$y = x + 1$  meets  $y = 5 - x$



## How do I do it?

## 1 Sketch

Draw the lines roughly; label points.

## 2 Intercepts

$x = 0$  and  $y = 0$ .

### 3 Meet

Set the  $y$ 's equal, solve for  $x$ , then  $y$ .

## 4 Measure

Area, length or midpoint as asked.

## SET THE $y$ 's EQUAL

**SET THE y's EQUAL** Intercepts:  $x = 0, y = 0$ .

**V5** Lines and Shapes

VARIANT 5 · SEE IT DONE, THEN TRY


  
Your notes
 WHEN YOU SEE → YOU DO

"The line crosses the axes at A and B.  
Find the area of AOB"

→ **Intercepts;  $\frac{1}{2} \times \text{base} \times \text{height}$** 

"The lines meet at P"

→ **Set equal; solve**

"perpendicular bisector of AB"

→ **Midpoint,  $-\frac{1}{m}$  (Topic 2)****WORKED EXAMPLE** DR ESAM

Lines and shapes · 3 marks

The line  $y = -\frac{1}{2}x + 3$  crosses the x-axis at A and the y-axis at B. O is the origin. Find the area of triangle AOB.

**STEP 1**

$x = 0 : B(0, 3)$

◀ y-intercept

**STEP 2**

$0 = -\frac{1}{2}x + 3 \Rightarrow A(6, 0)$

◀ x-intercept

**STEP 3**

$\frac{1}{2} \times 6 \times 3$

◀ right angle at O

**STEP 4**

$= 9$

◀ area

**ANSWER** 9 square units

✓ base 6, height 3 ✓

**WATCH OUT**

× x-intercept:  $x = -\frac{1}{2} \times 0 + 3$   
✓ put  $y = 0$

The x-axis is where  $y = 0$ .

**EXAM TIP**

A sketch with the intercepts marked earns method marks on its own.

**Check yourself**

calculator first · answers on page 15

**5.1**

Dr Eslam

The lines  $y = x + 1$  and  $y = 5 - x$  meet at P. Find P.

Answer: .....

**5.2**

Dr Eslam

Find the area of the triangle formed by  $y = 2x$ ,  $y = 6 - x$  and the x-axis.

Answer: .....

**5.3**

Dr Eslam

A(1, 2), B(5, 4). Find the perpendicular bisector of AB.

Answer: .....

## CHECK YOURSELF

# Answers



Every answer below was checked with sympy (check\_answers.py).

**1.1**  $-3, -1, 1, 3, 5$

**1.2**  $1, 3, 5, 7, 9$

**1.3**  $(-1, 10)$  and  $(4, 0)$

**2.1**  $m = 1.5, (0, -2)$

**2.2**  $(2, 0)$

**2.3** Yes: both  $m = 3$

**3.1**  $y = -\frac{1}{2}x + 1$

**3.2**  $y = 1.5x - 3$

**3.3**  $y = 2x + 1$

**4.1**  $\frac{5}{2}$

**4.2**  $y = 6 - 5x$

**4.3**  $y = -\frac{1}{3}x + 4$

**5.1**  $P(2, 3)$

**5.2**  $12$

**5.3**  $y = -2x + 9$

## THE WHOLE TOPIC ON ONE PAGE

# Linear graphs cheat sheet


  
Your notes
**V1 Draw 1/7**

$$y = 7 - 4x$$

Table, plot, rule.

**V2 Read  $m, c$** 

$$4x + 2y = 6 \rightarrow y = -2x + 3$$

 $y$  alone first.**V3 From a graph 3/7**

$$y = 1.5x - 1$$

 $c$  first, then rise over run.**V4 Parallel,  $\perp$  3/7**

$$-2 \rightarrow \frac{1}{2}$$

Same  $m$  or  $-\frac{1}{m}$ ; point gives  $c$ .**V5 Lines & shapes**

$$\frac{1}{2} \times 6 \times 3 = 9$$

Intercepts; set the  $y$ 's equal.**TOP 3 LOST MARKS**

1. reading  $m$  before  $y$  is alone  $\times$
2. counting squares, not the scale  $\times$
3. perpendicular: only the sign changed  $\times$

**BOTTOM LINE**  $y = mx + c$  ·  $y$  alone first · rise over run · parallel same, perpendicular  $-\frac{1}{m}$ .